

Tarun Mittal

Senior Software Engineer

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SUMMARY

- Senior Software Engineer with **5 years** owning cloud-native, high-throughput distributed systems end to end.
- Deep in **Python and Go microservices**, event-driven architectures, real-time and batch data processing, plus **Agentic AI and RAG**.
- Track record shipping production backends, leading migrations, and improving reliability, latency, and scale.

SKILLS

- Languages** – Python, Go (Golang), SQL, C, C++.
- Backend & APIs** – FastAPI, Gin, Django, SQLAlchemy, Pydantic, REST, gRPC, Celery, asyncio, Design Patterns.
- AI & Agents** – LLM agents (OpenAI/Gemini/Claude), LangGraph, LangChain, MCP, RAG, pgvector.
- Data & Streaming** – Kafka, Flink, Spark, ClickHouse, BigQuery, PostgreSQL, MySQL, Redis, MongoDB, Elasticsearch, DynamoDB, S3.
- Cloud & Infra** – GCP, AWS, Docker, Kubernetes, Linux, CI/CD, Bazel, Observability (ELK, New Relic, Datadog, PostHog, LangSmith).
- Core** – Distributed Systems, Microservices, System Design (HLD/LLD), Fault Tolerance, Concurrency, High Availability, Caching, Sharding, Performance Tuning, Mentorship.

PROFESSIONAL EXPERIENCE

Senior Software Engineer IMPACT ANALYTICS, Bangalore May 2026 – Present

Agentic AssortSmart (AI Powered Retail Merchandise Planning Platform)

- Raised planner throughput by targeting **20 to 100 cluster configs per plan vs 1** and turnaround **under 1 hour** (from days) by building the **Cluster Recommendation Copilot (FastAPI, LangGraph, MCP)** over a **Go doing layer**.
- Drove failure rate toward **under 2% from 8.5%** with **100% reproducible clusters** and **p95 probes under 500ms** vs 1 to 20s BigQuery via **read-only agent tools and 3 approval gates**.
- Unified manual REST and agent tool paths for Hindsight, Clustering, and Strategy in **Go (Gin)** with JWT auth and **Datadog, LangSmith, and PostHog** on a shared OTEL trace id.
- Cut **250M-row** pivot grids from **189s to 12.3s** (~15x) and avoided materializing **12B store-week rows (100 to 450x)** by adopting **ClickHouse/GCS** insert-only versioned planning storage.
- Tech** – Python, FastAPI, LangGraph, MCP, Go, Gin, ClickHouse, BigQuery, GCS, Datadog, LangSmith, PostHog, GCP, Docker.

Software Development Engineer 2 UBER (via EPAM Systems), Bangalore

July 2024 – May 2026

Financial Risk Management (FRM) Scoping Platform (Uber Finance)

- Owned Uber's **FRM Risk Scoping platform (FastAPI, MySQL, React)** across **8 screens and 30+ REST APIs**, automating quarterly scoping at **\$340M group materiality** and targeting a **70% cut** in manual reconciliation (~2 weeks to ~3-4 days).
- Designed an **11-table SQLAlchemy 2.0 MySQL** architecture with handler, service, and repository layers that auto-flags material items across **55 line items and 14 entities**.
- Owned Sheets to **MySQL v2 recon migration across 18 files**, validating HFM extracts against public **10-Q** filings and fixing a latent column-aliasing bug.
- Led 3 engineers** through design reviews, API contracts, and CI quality gates for releases feeding PwC audit work papers.
- Tech** – Python, FastAPI, SQLAlchemy 2.0, Pydantic, MySQL, Bazel, Docker.

Menu Ingestion & Automation Platform (Uber Eats)

- Cut menu onboarding from **24h to 2h** and saved **\$600K+ annually** on **30K+ menus/month** by shipping **Selenium** scrapers on GCP into **Kafka** (~200-500 peak events/sec) with **Flink** normalize/dedupe and **Spark** backfills.
- Hit **98% fidelity and 100% schema consistency** (offline eval) with a **RAG and Gemini 2.5 Pro** pipeline plus supervised fine-tuning.
- Raised successful ingestions **95%** by defeating anti-bot defenses with IP rotation and dynamic proxy pools.
- Automated **ANZ driver document compliance to 99.9%**, saving **20 hours/week** of manual verification.
- Tech** – Python, Selenium, Kafka, Flink, Spark, Gemini, RAG, SFT, GCP, Docker.

Software Development Engineer 2

MASTERS INDIA, Noida

December 2022 – June 2024

GST Compliance & E-Invoicing Platform

- Cut p95 latency from **1.2s to 300ms** for **1,500+ clients** by leading the **PHP monolith to FastAPI microservices** strangler migration and mentoring **2 engineers**.
- Scaled bulk e-invoicing to **100K+ per import and 1M+ daily** (~12 TPS avg, 100+ peak) and lifted throughput from **700 to 4,000 requests/min** (~67 RPS) via async Kafka and PostgreSQL quarter sharding.
- Cut redundant database reads **30%** with Redis caching while hardening bulk paths with **idempotency keys, retries, and DLQ replay**.
- Cut incident triage **70%** with **ELK and New Relic** on-call alerting and raised coverage from **35% to 82%** at **98%** deployment success.
- Tech** – Python, FastAPI, Kafka, PostgreSQL, MongoDB, Redis, Elasticsearch, Celery, Docker, ELK, New Relic, AWS.

Software Development Engineer

GEEKSFORGEEKS, Noida

August 2021 – November 2022

- Improved reliability for **10K+ daily queries** and **10x contest spikes** by migrating the doubt-support backend from **PHP to Django (Python)**.
- Lifted **premium subscriptions 15 to 20%** by designing voting and pinning APIs on **MySQL, MongoDB, Redis, and Elasticsearch**.
- Raised **course sales 30%** and operational efficiency **70%** by launching an influencer analytics dashboard and cron pipelines for video processing.

EDUCATION

B.Tech (Information Technology)

Institute of Engineering & Technology, Lucknow

July 2017 – June 2021

ACHIEVEMENTS & CERTIFICATIONS

- Ranked **2260 / 37,000+** in the Google Code Jam Qualifier 2021 and finalist (top 3 nationally) at the **Smart India Hackathon 2020 (MHRD)**.
- Participated in the **Global AI Hackathon** at EPAM Systems.
- Earned **HackerRank Problem Solving** and **LangChain Academy** certifications ([HackerRank](#), [LangChain](#)).